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General Comment

this is an updated version of my comment - I just noticed that the RFI requested 12 point font so I have updated the formatting to comply with that request

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Attachments

Ron Bodkin Input on NSF AI Action Plan

Response to Request for Information: Development of an Artificial Intelligence (AI) Action Plan

Respondent Information

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Introduction

As the Founder and CEO of ChainML, which develops Theoriq - an Artificial Intelligence (AI) agent swarm protocol enabling the agentic economy, I am pleased to submit this response to the Request for Information regarding the Development of an Artificial Intelligence Action Plan. I bring perspectives from multiple leadership roles across the AI ecosystem: previously serving as VP of AI Engineering and CIO at Vector Institute where I grew the technology team to 30 people and supported 500 researchers with computing infrastructure and engineering; leading Applied AI and Responsible AI efforts in the Google Cloud CTO office where I spearheaded collaborative innovation efforts with strategic customers and Google AI research and product teams; founding CEO of enterprise AI/ML startup Think Big Analytics (acquired by Teradata); VP Engineering at Quantcast leading data science and engineering teams applying AI for real-time advertising; and Co-Founder and CTO of C-Bridge Internet Solutions. My educational background includes an honors B.S. in Math and Computer Science from McGill University and a Master's in Computer Science from MIT.

This response focuses on the critical importance of prioritizing AI alignment and control research as a foundational element of the AI Action Plan. This recommendation directly supports the stated goals of sustaining and enhancing America's AI dominance while avoiding unnecessarily burdensome regulations. Rather than restricting private sector innovation, investment in alignment research will accelerate it by building the technical foundations that enable safe, reliable, and trustworthy AI systems at scale.

Defining AI Alignment and Cooperative AI Research

AI alignment research focuses on ensuring that AI systems reliably pursue the goals and values intended by their creators, even as they become more capable and potentially autonomous. This includes developing techniques to make AI systems interpretable, robust to distribution shifts, resistant to adversarial manipulation, and aligned with human preferences and values [1].

Cooperative AI, a related research area, focuses on developing AI systems that can effectively cooperate with humans and other AI systems, addressing challenges like coordination, communication, and negotiation in mixed human-AI environments [2]. As AI agents are increasingly deployed in the economy, the dynamics of AI interactions and mechanisms to ensure effective collaboration and avoid perverse incentives will be of great importance. Together, these research directions form a crucial foundation for ensuring that increasingly powerful AI systems remain beneficial, controllable, and safe.

AI Alignment and Cooperative AI Research: A Strategic Priority for U.S. Leadership

National Security and Innovation Leadership

AI alignment and cooperative AI research represent critical infrastructure for maintaining U.S. leadership in artificial intelligence. As AI systems become more capable and autonomous, ensuring they reliably pursue intended goals becomes increasingly important. Alignment failures in critical systems could create significant national security vulnerabilities as well as limit economic opportunity from scaling deployment. Conversely, strong alignment capabilities create competitive advantages for US AI systems in global markets.

Of particular importance is research into tamper-resistant training methods that allow for the safe release of powerful open-source models even in the face of fine-tuning by adversaries. Such methods can mitigate risks of abuse—such as CBRN (Chemical, Biological, Radiological, Nuclear) attacks [3] or use by adversaries for military applications—while still enabling the vibrant research ecosystem that has been instrumental to American AI leadership. This balance represents a nuanced approach that promotes innovation while addressing legitimate security concerns.

This approach aligns with existing NSF initiatives such as the National AI Research Institutes program, which has already established centers focused on trustworthy AI to ensure systems are "safe, secure, fair, transparent and accountable." By strengthening our focus on alignment and cooperative AI, we can build upon this foundation to ensure the United States leads not only in AI capabilities but also in AI safety—a strategic necessity in the global AI landscape where other nations are rapidly advancing their own AI governance frameworks.

As noted by several RFI respondents, including Anthropic, advanced AI systems with capabilities exceeding human experts in most disciplines are likely to emerge within a decade. Rather than imposing preemptive restrictions that could hamper innovation, as some have suggested, investing in alignment and cooperative AI research provides the scientific foundation needed to develop these systems responsibly while maintaining American leadership.

Economic Competitiveness

Reliability, trustworthiness, and accuracy are not merely desirable features of AI systems—they are prerequisites for scaled investment and enterprise adoption. Organizations will only commit significant resources to AI implementation when they can trust systems to perform reliably across diverse operating conditions. Alignment research directly addresses these requirements by developing techniques to ensure AI systems behave as intended even as they become more capable and autonomous.

The concept of "defensive acceleration" is critical here: investing in differentiated AI capabilities like superior alignment is not merely defensive but represents a strategic advantage. U.S. leadership in AI alignment research will create cascading competitive advantages throughout the AI supply chain, as systems with superior reliability and alignment characteristics will be required for many applications.

Perhaps most significantly, science itself will be massively accelerated with reliable AI systems. AI can dramatically increase the pace of scientific discovery when researchers can trust its outputs and analysis, justifying a significant reallocation of NSF budget funds from other priorities to AI alignment and cooperation. This represents an investment multiplier effect, as improvements in AI alignment will accelerate progress across virtually all scientific domains.

This economic imperative is reinforced by international developments. The European Union has positioned "excellence and trust" as twin pillars of its AI strategy [4]. Similarly, China has emphasized creating AI that is "safe, reliable, controllable and equitable" in its national strategy [5]. For the United States to maintain economic competitiveness in AI, exceeding these investments in alignment and cooperative AI is essential—not only to create better AI systems but also to shape global norms and standards that align with American values and interests.

This approach addresses concerns raised by industry stakeholders about maintaining American competitiveness and fostering innovation. Unlike restrictive regulations that could hamper progress, investing in alignment research enhances the usability, reliability, and trustworthiness of AI systems—qualities that will accelerate adoption and economic benefits. As many industry groups have emphasized, America's AI leadership depends on creating an environment where innovation flourishes. By funding fundamental research in alignment and cooperative AI, we create the technical foundation for both innovation and safety, addressing legitimate concerns without imposing unnecessary burdens on developers.

Public Good Research Justification

Market incentives alone are likely to underinvest in alignment research for several reasons:

1. The benefits are widely distributed across the entire AI ecosystem
2. Commercial timelines may prioritize short-term capabilities over long-term safety
3. The most critical alignment challenges affect advanced systems that are still in development

This classic public goods problem justifies government funding of foundational alignment research. We already have strong evidence that such investments yield tremendous returns. Several key results from AI alignment research have proven crucial to unlocking economic value and U.S. leadership:

- **Reinforcement Learning from Human Feedback (RLHF):** Building on rich antecedents in the academic research, RLHF [6] has become the industry standard for aligning large language models with human preferences, enabling systems like ChatGPT that have created billions of dollars in economic value.
- **Constitutional AI:** Research into automated AI alignment techniques like Constitutional AI [7] has enabled more reliable and useful AI assistants like Claude, providing robust frameworks for encoding human values and preventing harmful outputs.
- **Mechanistic Interpretability:** Advances in understanding model internals have improved debugging, enhanced security, and enabled more targeted model improvements. Specific breakthroughs include circuit analysis in models [8], identification of factual knowledge in language models [9], and techniques for detecting hidden capabilities or deceptive behavior in models [10].

- **Cooperative AI:** Research on multi-agent systems that can effectively collaborate with humans and other AI systems, addressing challenges in areas like communication protocols and alignment of incentives [11]

These examples demonstrate how alignment research directly enhances capabilities rather than constraining them. They represent cases where the technical foundations laid by public research enabled dramatic commercial innovations.

Research Priority Areas

To maintain U.S. leadership in AI alignment and cooperative AI, I recommend prioritizing the following research areas:

1. **Advanced Interpretability and Explainability:** Developing techniques to understand the internal functioning of increasingly complex AI systems, enabling more effective oversight and targeted improvements. This aligns with what NSF has identified as "critically important to further advance our understanding of AI, including aspects of transparency, security, and control" [12].
2. **Scalable Oversight Methods:** Creating techniques that allow meaningful human oversight of AI systems as they become more complex and operate at greater speeds. This addresses the fundamental challenge of ensuring AI remains under human control, a concern shared globally.
3. **Robustness Against Adversarial Attacks:** Strengthening AI systems against both intentional manipulation and unintentional distribution shifts, a key component of the science of understanding AI that enables us to anticipate and prevent failures.
4. **Long-term Alignment Challenges:** Research addressing the technical challenges of ensuring increasingly capable systems remain aligned with human values and intentions, ensuring that as AI models become more autonomous, their objectives and behaviors remain consistent with the ethical, factual, and safety expectations set by humans.
5. **Tamper-Resistant Training:** Developing methods that ensure advanced AI systems cannot be easily modified to remove safety constraints while still enabling valuable research with open models. For example, Circuit Breakers [13] is an innovative technique to make open weight systems resistant to fine-tuning attacks.
6. **Cooperative AI Systems:** Designing AI that can work effectively in teams with humans and other AI systems, follow our instructions in spirit, negotiate and resolve conflicts, and generally behave as reliable partners to humanity and work well together to further our ends while mitigating negative externalities. This

emerging discipline focuses on AI that can engage in collaborative problem-solving rather than purely competitive or black-box behaviors.

Policy Recommendations

Funding Mechanisms

1. **Dedicated AI Alignment and Cooperative AI Research Program:** Establish a dedicated program within NSF specifically for AI alignment and cooperative AI research, with a significant budget allocation (recommended minimum: $2 billion annually). This represents an appropriate scale of investment given the foundational importance of this research to AI progress and safety.
2. **Multi-Agency Coordination:** Create an interagency working group to coordinate alignment and cooperative AI research across NSF, DARPA, DOE, and other relevant agencies.
3. **Public-Private Matching Grants:** Implement matching grant programs where government funding is paired with private sector contributions, incentivizing industry investment in alignment and cooperative AI research.
4. **Diverse Research Ecosystem:** Ensure funding is distributed across leading universities, independent research organizations (such as non-profits focused on AI safety), and open-source development communities. This diversity of approaches is crucial for tackling the multi-faceted challenges of alignment.

Research Infrastructure

1. **National Research Computing Infrastructure:** Significantly expand investment in the National Artificial Intelligence Research Resource (NAIRR) program, prioritizing access for alignment and cooperative AI research during the pilot phase and accelerating its scaling. Given the compute-intensive nature of modern AI research, dedicated computing allocations are essential for experiments requiring large-scale models [14]
2. **Data Resources for Evaluation:** Develop standardized benchmark datasets and evaluation frameworks for assessing progress in alignment and cooperative AI research, in collaboration with the US AI Safety Institute.
3. **Coordination with AI Safety Institute:** Establish formal feedback channels between the U.S. AI Safety Institute and research funding agencies to ensure that industry learnings about emerging risks and challenges directly inform research priorities.

Education and Workforce Development

1. **AI Alignment Curriculum Development:** Fund the development of educational materials on AI alignment and cooperative AI for undergraduate and graduate programs.
2. **Fellowships and Scholarships:** Create dedicated fellowship programs for students pursuing research in AI alignment and cooperative AI.
3. **Industry-Academic Rotations:** Facilitate researcher exchanges between industry and academia to cross-pollinate ideas and approaches.

Complementary to Broader AI Policy Priorities

It is important to recognize that investment in AI alignment and cooperative AI research should complement, not replace, other important policy priorities. Many RFI respondents have rightfully highlighted concerns about algorithmic bias, privacy, labor displacement, and other immediate impacts of AI systems. These issues deserve serious attention and should be addressed through appropriate policy mechanisms.

The research agenda proposed here is complementary with efforts to address these concerns. Improved interpretability techniques, for example, can help detect and mitigate bias in AI systems. Better cooperative AI approaches can enable more equitable human-AI collaboration that preserves human agency and enhances worker productivity rather than displacing jobs. And as suggested by several respondents, including advocates for near-term AI ethics, addressing both current and future risks is essential for fostering public trust in AI technologies.

Furthermore, the technical foundations developed through alignment and cooperative AI research will provide the tools and insights needed for effective governance as AI systems continue to advance. Rather than implementing premature regulatory measures based on incomplete understanding of AI systems, this research-first approach creates the knowledge base that can inform targeted, effective policy interventions when necessary.

In reviewing the diverse perspectives submitted to this RFI, I recognize that stakeholders have proposed a wide range of approaches to AI governance. Some emphasize immediate regulatory interventions, including mandatory "off-switches" or strict controls on frontier AI development. Others caution against overemphasizing speculative risks and advocate for light-touch frameworks that prioritize innovation and competitiveness. Industry associations have largely favored voluntary standards and self-regulation, while civil society organizations often prioritize addressing current harms like algorithmic bias and privacy concerns.

This response acknowledges these varying perspectives while advocating for a balanced approach: targeted government investment in AI alignment and cooperative AI research represents a prudent middle path. Rather than imposing heavy-handed regulation that could stifle innovation or dismissing legitimate concerns about increasingly capable AI systems, building US research capacity to understand and control AI is the right focus for government investment now.

By strengthening our scientific understanding of AI systems and developing techniques to ensure their reliability, we create the foundation for both innovation and safety. This approach addresses concerns raised by companies like Anthropic in their response to this RFI about the development timeline for powerful AI systems, without resorting to the more restrictive measures suggested by organizations like the Future of Life Institute in their response, where there is not yet evidence of near term risks that warrant such restrictions.

I also support expanding investment in the AI Safety Institute and NIST more broadly to build skills in government, to support sensitive AI considerations that impact national security, and to provide measurement and coordination. I believe that these efforts are important and deserving of more funding but there is a major and critical cap in funding AI research for alignment and coordination that should be a top priority for the NSF AI Action Plan.

International Context and Strategic Implications

The United States is not alone in recognizing the importance of AI safety, alignment, and interpretability. Both allies and competitors are making substantial investments in these areas:

The **European Union** has positioned "excellence and trust" as twin pillars of its AI strategy, finalizing a comprehensive AI Act that imposes strict safety and transparency requirements on high-risk AI applications [4]. European research consortia are actively advancing explainable AI methods, bias reduction, and validation techniques under the banner of "Trustworthy AI."

Meanwhile, **China** has announced bold plans to be the global leader in AI by 2030, emphasizing building AI that is "safe, reliable, and controllable" [5]. This has led to significant government-backed research in AI robustness and new regulations on algorithmic transparency and generative AI oversight. China's Global AI Governance Initiative and AI Safety Governance Framework mirror Western principles, calling for safety, transparency, and accountability in AI development.

For the United States to maintain technological leadership and shape global norms, we must not only match but exceed these international efforts. By leveraging our strengths—world-class universities, thriving private-sector AI labs, and institutions like NSF—we can lead in AI alignment and cooperative AI while establishing standards that reflect democratic values.

Conclusion: Building Bridges Through Research Investment

Investment in AI alignment and cooperative AI research represents a strategic opportunity to enhance U.S. leadership in artificial intelligence while addressing legitimate safety concerns without imposing burdensome regulations. By creating the technical foundations for reliable, trustworthy AI systems, such research will accelerate rather than constrain innovation.

The recent examples of RLHF, constitutional AI, mechanistic interpretability, and cooperative AI demonstrate how this research directly contributes to capabilities and economic value. A significant allocation of resources to this area—on the order of billions of dollars annually through both direct funding and computing resources—would yield tremendous returns through both direct advances in AI technology and indirect acceleration of scientific progress across domains.

Building upon existing NSF initiatives like the National AI Research Institutes, NAIRR, and coordination with the AI Safety Institute, we can establish a comprehensive national strategy for AI alignment and cooperative AI research. This approach will not only protect our society from AI-related risks but also position the United States to lead by example—demonstrating that cutting-edge AI innovation can go hand-in-hand with robust safety measures.

This research-focused approach addresses concerns from across the spectrum of perspectives represented in RFI responses. For those concerned about potential risks from advanced AI systems, it provides concrete mechanisms to understand and mitigate those risks. For industry stakeholders prioritizing innovation and competitiveness, it builds the foundation for more capable, reliable AI that can be deployed with confidence. And for those focused on near-term impacts, it develops tools that can help address current challenges while preparing for future advances.

The United States has an opportunity to lead the world in establishing not just the most advanced AI capabilities, but also the most reliable, trustworthy, and beneficial systems. This leadership position will create cascading advantages throughout our economy and society while fostering international partnerships that advance AI for human flourishing.

I appreciate the opportunity to provide this input to the AI Action Plan and would welcome further discussion on these recommendations.

Respectfully submitted,

Ron Bodkin
Founder and CEO, ChainML

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